

Physical Chemistry of Materials

Lecture 20

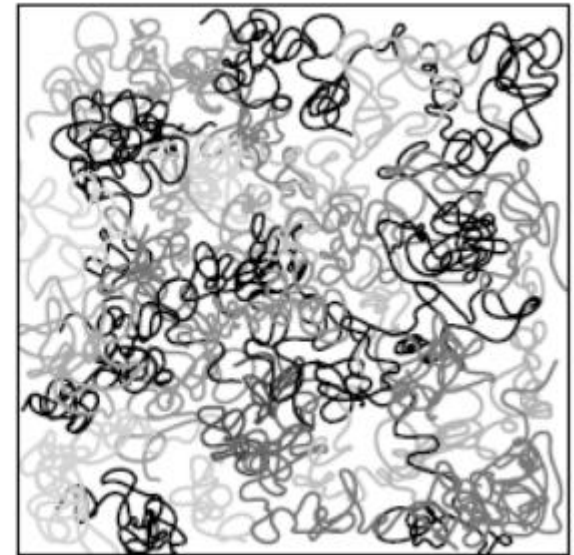
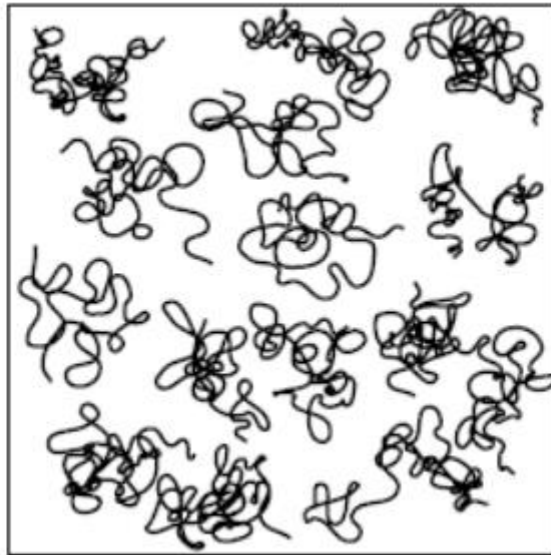
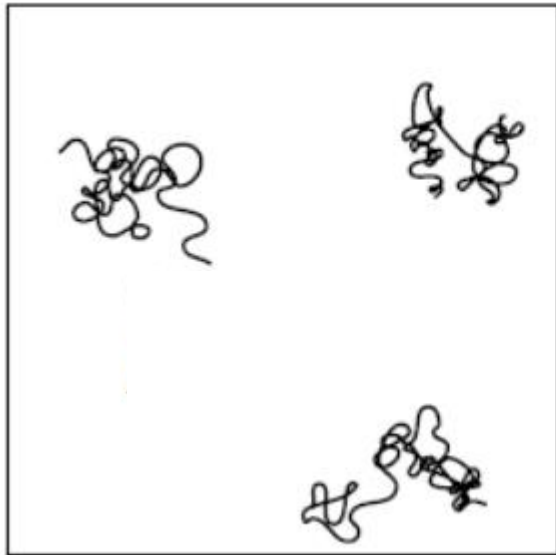
Real Chains Polymer Solutions



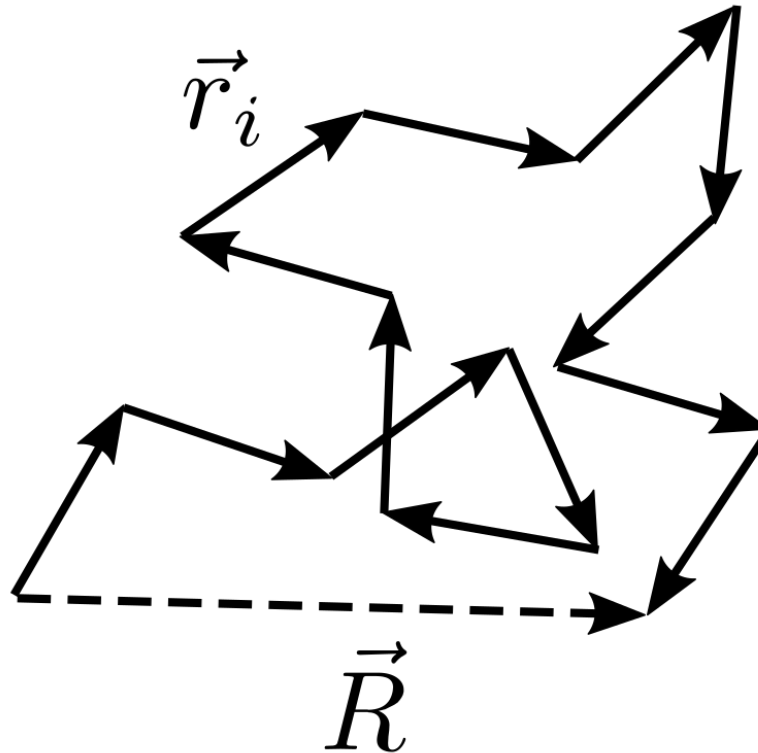
Last Class

Polymer Solutions

polymer concentration



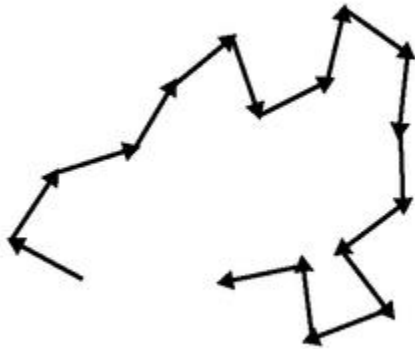
Ideal Chains



(Image: https://commons.wikimedia.org/wiki/File:Ideal_chain_random_walk.svg)

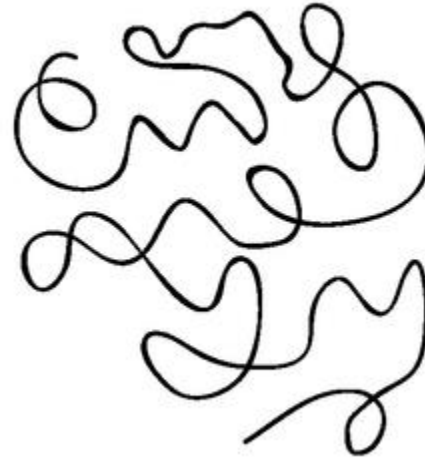
Models for Ideal Chains

A



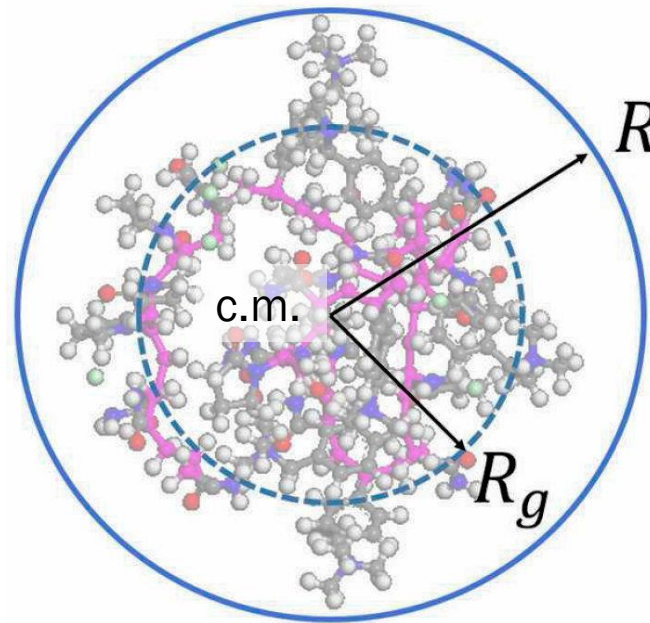
Freely jointed chain (FJC)

B



Worm-like chain (WLC)

Radius of Gyration R_g



$$R_g^2 = \frac{1}{N} \sum_i^N (\vec{x}_i - \vec{x}_{\text{cm}})^2 \quad (\vec{r}_i = \vec{x}_i - \vec{x}_{i-1})$$

Radius of Gyration R_g

$$R_g^2 = \frac{1}{N} \sum_i^N (\vec{x}_i - \vec{x}_{\text{cm}})^2$$

As $\vec{x}_{\text{cm}} = \frac{1}{N} \sum_i^N \vec{x}_i$, one can show that

$$R_g^2 = \frac{1}{N^2} \sum_{i,j}^{N,N} (\vec{x}_i^2 - \vec{x}_i \cdot \vec{x}_j) = \frac{1}{2N^2} \sum_{i,j}^{N,N} (\vec{x}_i - \vec{x}_j)^2$$

$$\langle R_g^2 \rangle = \frac{1}{2N^2} \sum_{i,j}^{N,N} \langle (\vec{x}_i - \vec{x}_j)^2 \rangle$$

R_g of FJC

$$\langle R_g^2 \rangle = \frac{1}{2N^2} \sum_{i,j}^{N,N} \langle (\vec{x}_i - \vec{x}_j)^2 \rangle = \frac{1}{N^2} \sum_{i=1}^N \sum_{j=i}^N \langle (\vec{x}_i - \vec{x}_j)^2 \rangle$$

Using the end-to-end distance $\langle (\vec{x}_i - \vec{x}_j)^2 \rangle = (j - i)a^2$,

$$\sum_{i=1}^N \sum_{j=i}^N (j - i) \approx \int_0^N \int_u^N (v - u) dv du = \frac{N^3}{6}$$

$$\langle R_g^2 \rangle = \frac{Na^2}{6} \propto N$$

Power Laws

For a freely jointed chain,

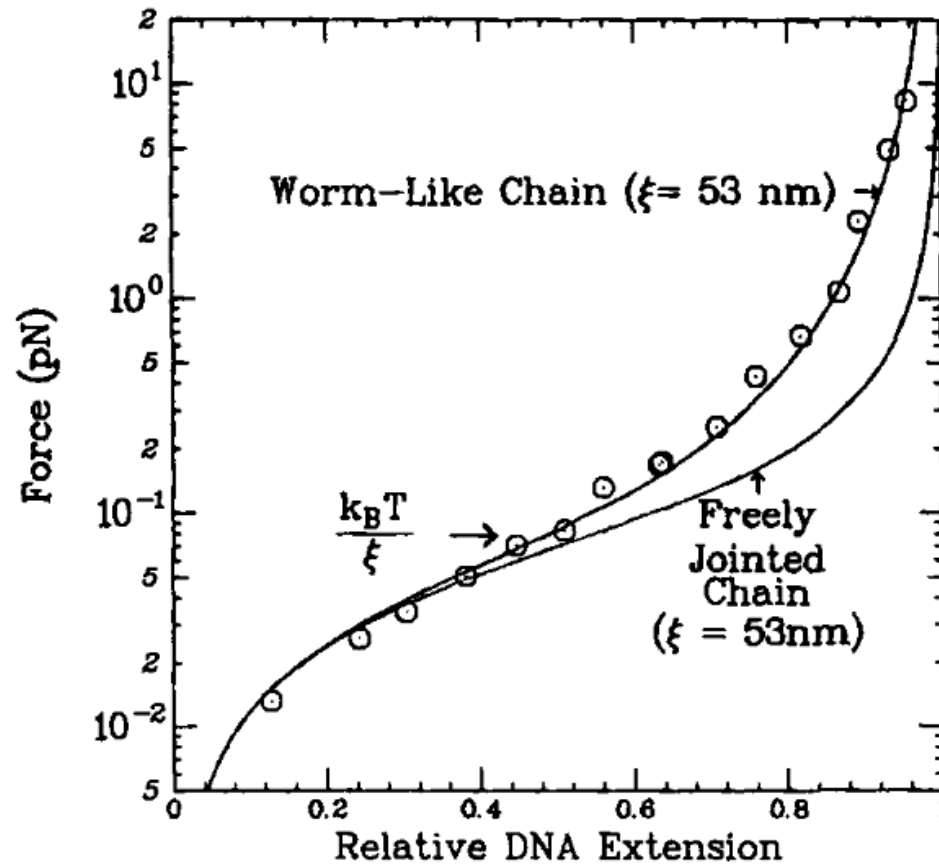
- End-to-end distance

$$\langle \vec{R}^2 \rangle^{1/2} = (Na^2)^{1/2} \propto N^{1/2}$$

- Radius of gyration

$$\langle R_g^2 \rangle^{1/2} = \left(\frac{Na^2}{6} \right)^{1/2} \propto N^{1/2}$$

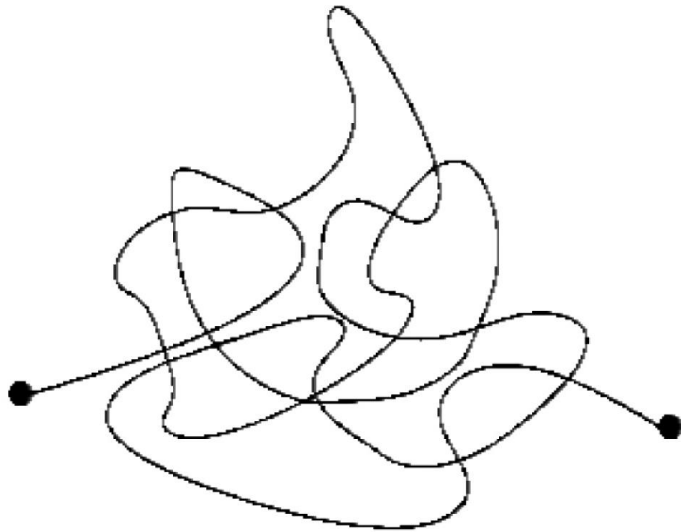
Force-Extension Curves



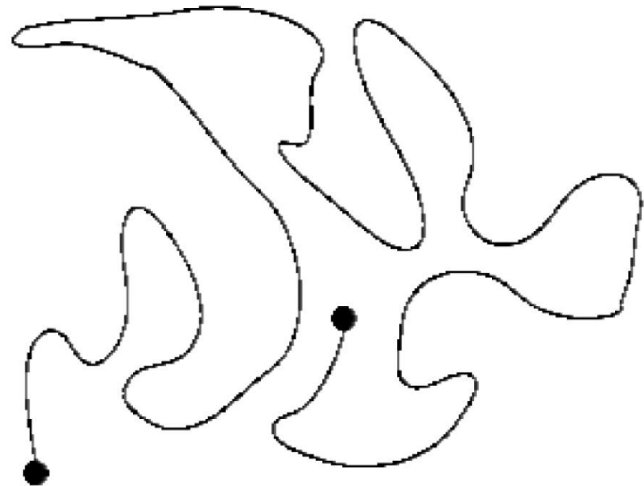
(Image: Strick et al., *J Stat Phys* (1998))

Long-Range Interactions

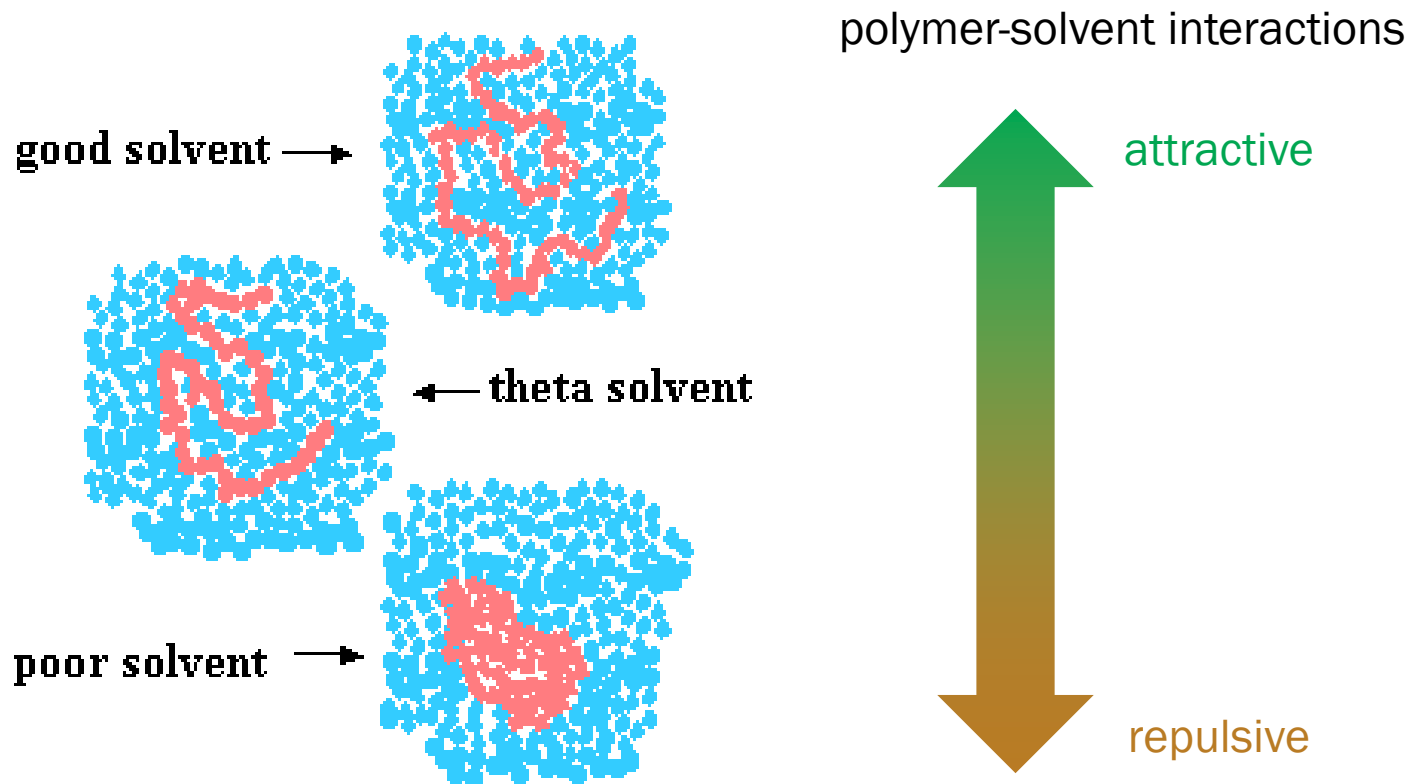
Ideal chains



“Real” chains



Origin of Long-Range Interactions



(Image: http://rkt.chem.ox.ac.uk/lectures/liqsolns/polymer_solutions.html)

Power Laws

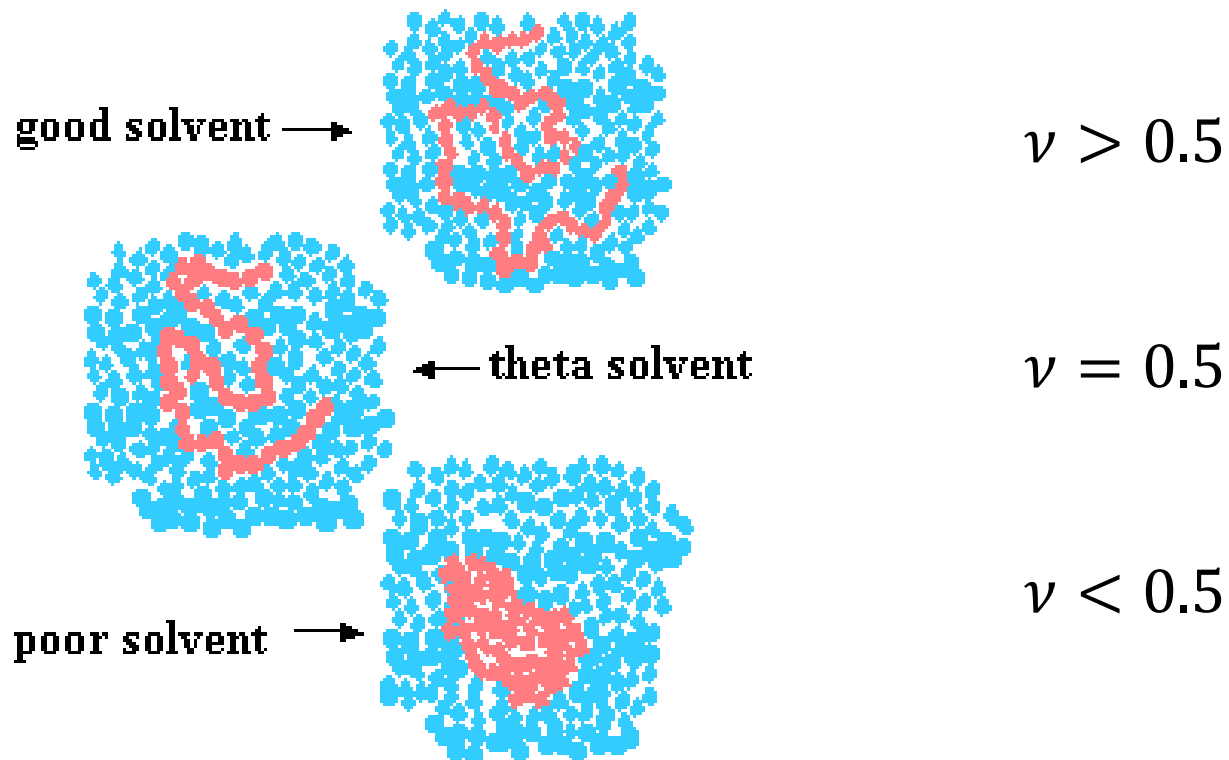
For a freely jointed chain,

$$\langle \vec{R}^2 \rangle^{1/2} = (Na^2)^{1/2} \propto N^{1/2}$$

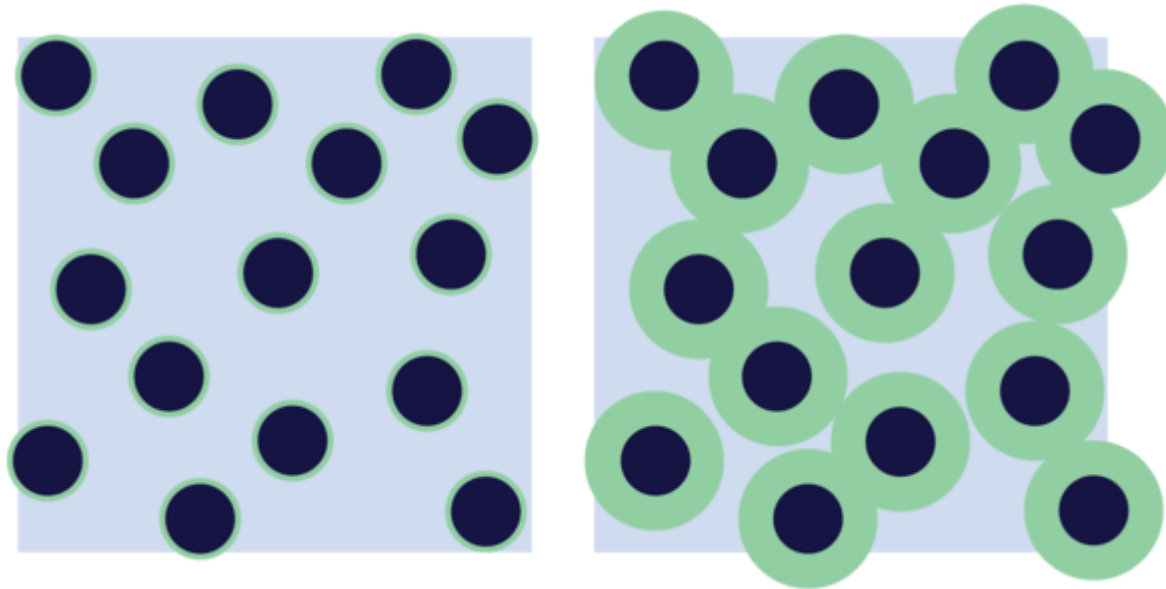
$$\langle R_g^2 \rangle^{1/2} = \left(\frac{Na^2}{6} \right)^{1/2} \propto N^{1/2}$$

Generally, $R \propto N^\nu$ (ν : Flory exponent)

Flory Exponents for Real Chains



Excluded Volume



Repulsion: positive excluded volume

Attraction: negative excluded volume

Flory Theory: Good Solvent

- Polymer with N monomers
- Each monomer occupies excluded volume of v_0
- Assume the polymer size $R \rightarrow$ volume $\sim R^3$
- Excluded-volume cost: $A_{\text{ev}} \approx k_B T \times v_0 (N/R^3) \times N$
- Entropic cost: $A_{\text{ent}} \approx k_B T R^2 / N a^2$ (blob argument)
- Free energy

$$A = A_{\text{ev}} + A_{\text{ent}} = k_B T \left(v_0 \frac{N^2}{R^3} + \frac{R^2}{N a^2} \right)$$

Minimization

$$A = A_{\text{ev}} + A_{\text{ent}} = k_{\text{B}}T \left(v_0 \frac{N^2}{R^3} + \frac{R^2}{Na^2} \right)$$

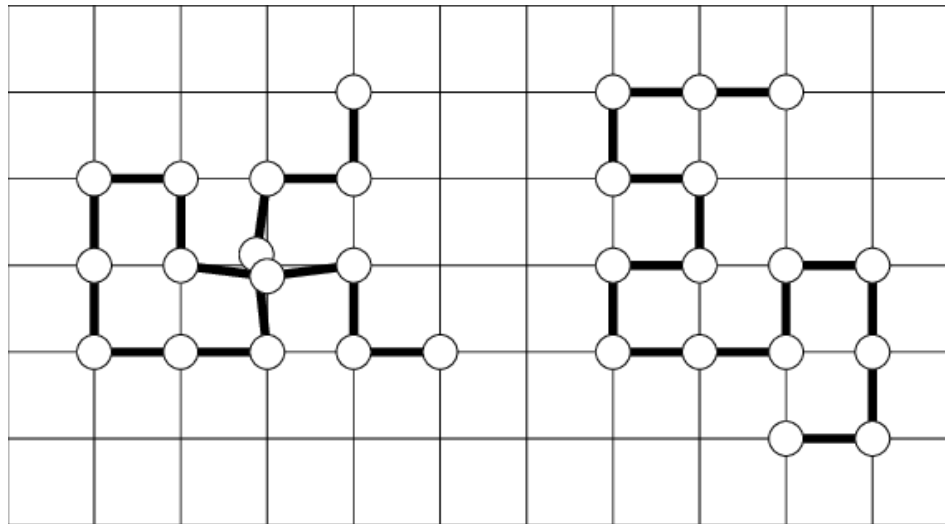
Minimizing free energy with respect to R ,

$$\frac{dA}{dR} = k_{\text{B}}T \left(-3v_0 \frac{N^2}{R^4} + 2 \frac{R}{Na^2} \right) = 0$$

$$R_{\text{good}} = v_0^{1/5} a^{2/5} N^{3/5} \propto N^{3/5}$$

Accurate Model: Self-Avoiding Walk

(de Gennes, 1972)



$$R_{\text{good}} \approx v_0^{2\nu-1} a^{4-6\nu} N^\nu$$

$$\nu \approx 0.588$$

Flory Theory: Poor Solvent

$$A = A_{\text{ev}} + A_{\text{ent}} = k_{\text{B}}T \left(v_0 \frac{N^2}{R^3} + \frac{R^2}{Na^2} \right)$$

- Poor solvent: $v_0 < 0 \rightarrow$ minimization gives $R \approx 0$
- At the dense limit,

$$\frac{A_{\text{ev}}}{R^3} \approx k_{\text{B}}T(v_0 c^2 + w c^3 + \dots)$$

Considering the three-body interaction with $w > 0$,

$$A = k_{\text{B}}T \left(v_0 \frac{N^2}{R^3} + w \frac{N^3}{R^6} + \frac{R^2}{Na^2} \right)$$

Minimization

$$A = k_{\text{B}}T \left(v_0 \frac{N^2}{R^3} + w \frac{N^3}{R^6} + \frac{R^2}{Na^2} \right)$$

For $R \ll \sqrt{Na}$,

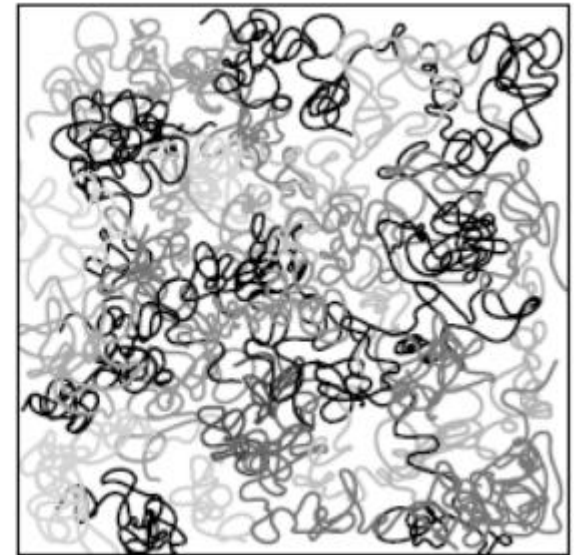
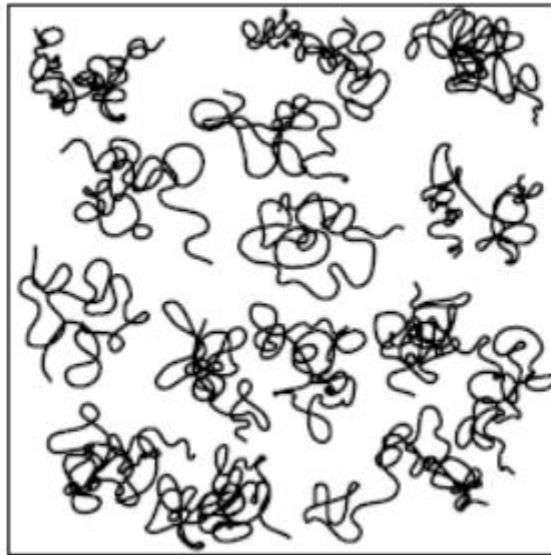
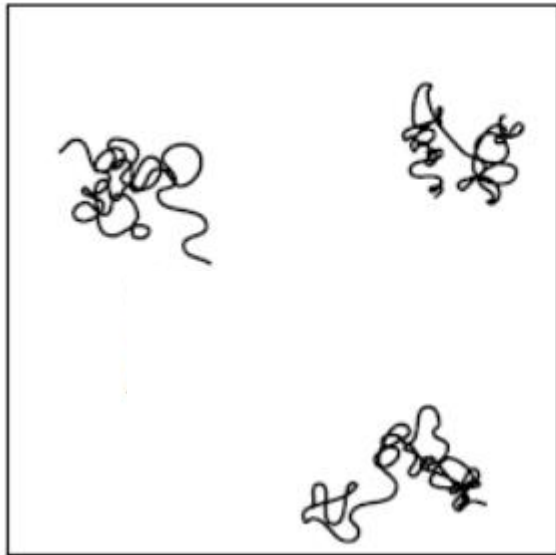
$$A \approx k_{\text{B}}T \left(v_0 \frac{N^2}{R^3} + w \frac{N^3}{R^6} \right)$$

Minimizing free energy with respect to R ,

$$R_{\text{poor}} = \left(\frac{wN}{|v_0|} \right)^{1/3} \propto N^{1/3}$$

Polymer Solutions

polymer concentration



“overlap”

Theta Solvent

$$R \sim a\sqrt{N} \text{ (at all concentrations)}$$

- Overlap volume fraction ϕ^*

(system volume fraction) = (volume fraction in a coil)

$$\phi^* = \frac{Na^3}{R^3} = \frac{1}{\sqrt{N}}$$

- Dilute solution: $\phi < \phi^*$
- Semidilute solution: $\phi > \phi^*$ (but still $\phi \ll 1$)

Good Solvent

$$R_{\text{good}} \approx v_0^{2\nu-1} a^{4-6\nu} N^\nu$$

- Overlap volume fraction ϕ^*

$$\phi^* = \frac{Na^3}{R_{\text{good}}^3} = \left(\frac{a^3}{v_0}\right)^{6\nu-3} N^{1-3\nu} \sim N^{-0.76}$$

What happens if $\phi > \phi^*$?

Thermal Blob

attraction < thermal energy

- Nearly ideal behavior: $\xi_T = a g_T^{1/2}$
- Flory interaction energy $A_{ev} \approx k_B T \times v_0 (N/R^3) \times N$

$$k_B T v_0 \frac{g_T^2}{\xi_T^3} \approx k_B T$$

$$\xi_T \approx \frac{a^4}{v_0}$$

Correlation Volume

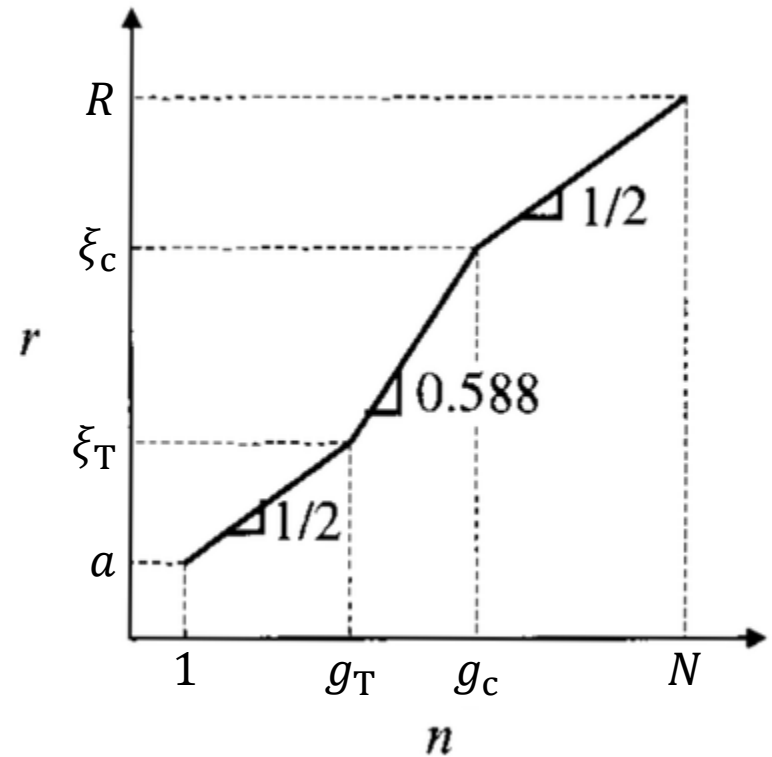
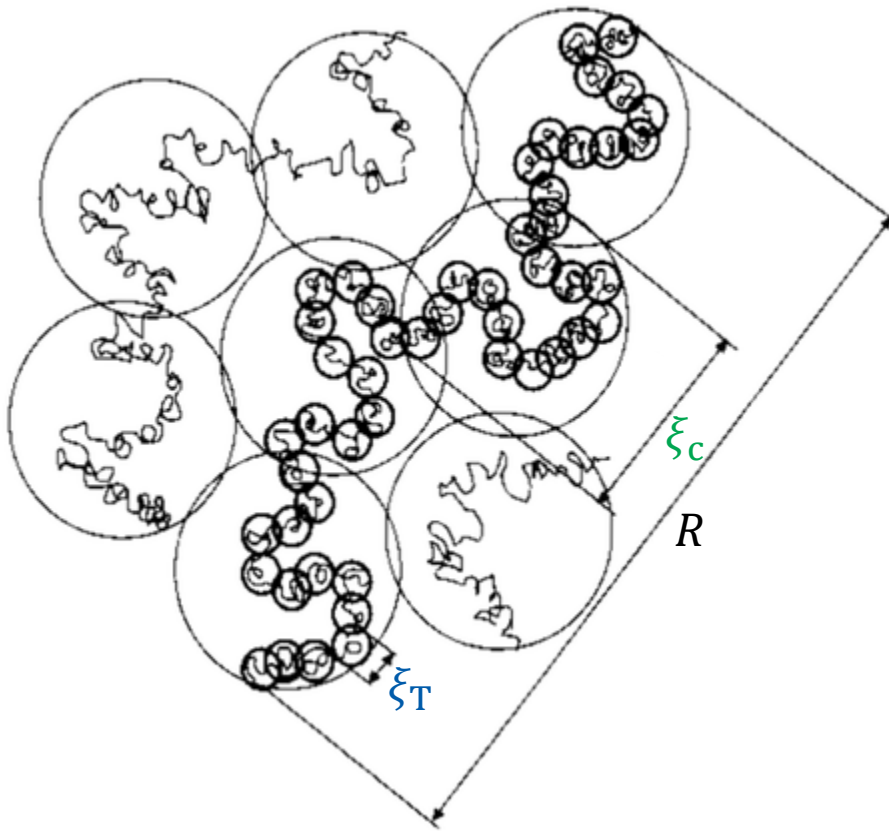
Attraction is NOT screened

- Good-solvent behavior: $\xi_c = v_0^{2\nu-1} a^{4-6\nu} g_c^\nu$
- Correlation volumes are space-filling

$$\phi = \frac{g_c a^3}{\xi_c^3}$$

$$\xi_c \approx a \left(\frac{a^3}{v_0} \right)^{(2\nu-1)/(3\nu-1)} \phi^{-\nu/(3\nu-1)}$$

Three Regimes



(Image: Rubinstein and Colby, *Polymer Physics* (2003), Fig. 5.5)

What If $\xi_T \geq \xi_c$?

$$\xi_T \approx \frac{a^4}{v_0}; \quad \xi_c \approx a \left(\frac{a^3}{v_0} \right)^{(2\nu-1)/(3\nu-1)} \phi^{-\nu/(3\nu-1)}$$

The crossover point is

$$\phi^{**} = \frac{v_0}{a^3}$$

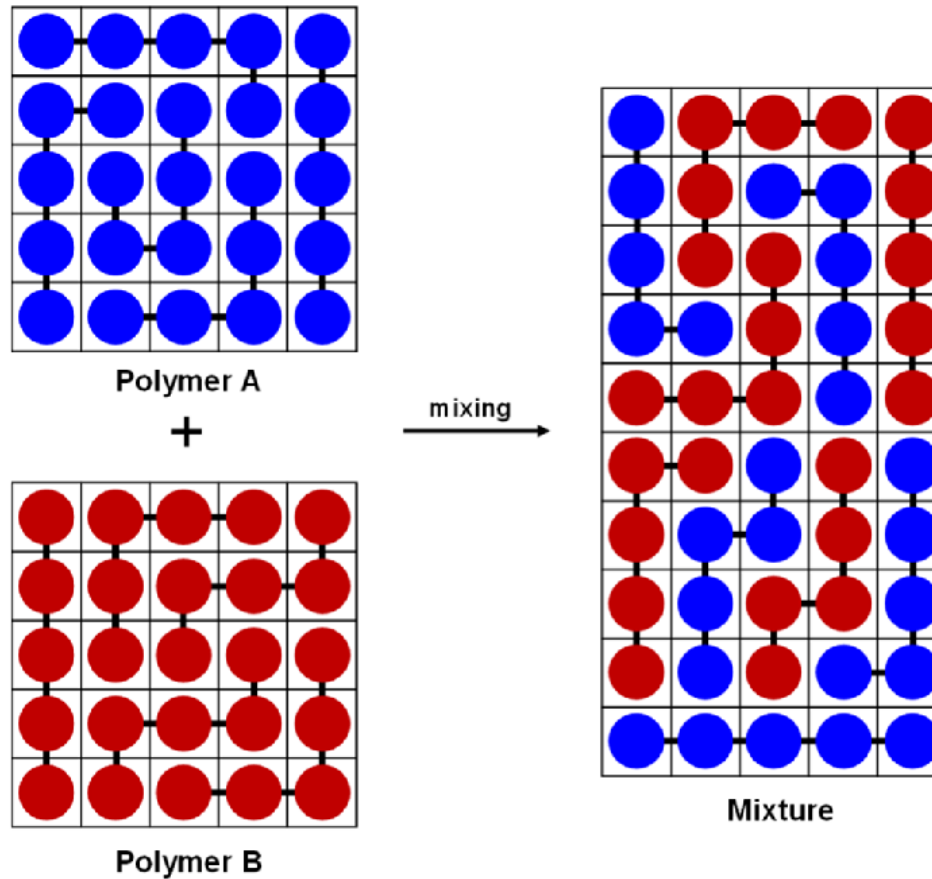
For $\phi > \phi^{**}$, chains are **ideal on all length scales**.

→ concentrated solution

Poor Solvent

Phase Separation!

Flory-Huggins Model



(Image: <https://link.springer.com/article/10.1007/s13367-023-00076-9>)

System Setup

For species A and B,

- Number of monomers in each polymer: N_A and N_B
- Volumes: V_A and V_B
- Volume fractions: $\phi_i = V_i / (V_A + V_B)$
- Total number of sites: $M = (V_A + V_B) / v_0$
 - Number of sites occupied by species i : $V_i / v_0 = M \phi_i$
- Number of polymers: $M_i = M \phi_i / N_i$

(Translational) Entropy

$$S = k_B \ln W$$

For each polymer of species i ,

- Mixed state: $W_{A+B} = M$
- Separated state: $W_i = M_i = M\phi_i$
- Entropy change upon mixing

$$\Delta S_i = S_{A+B} - S_i = k_B \ln \left(\frac{W_{A+B}}{W_i} \right) = -k_B \ln \phi_i$$

Entropy of Mixing

$$\Delta S_i = -k_B \ln \phi_i \text{ (per polymer)}$$

For the whole system,

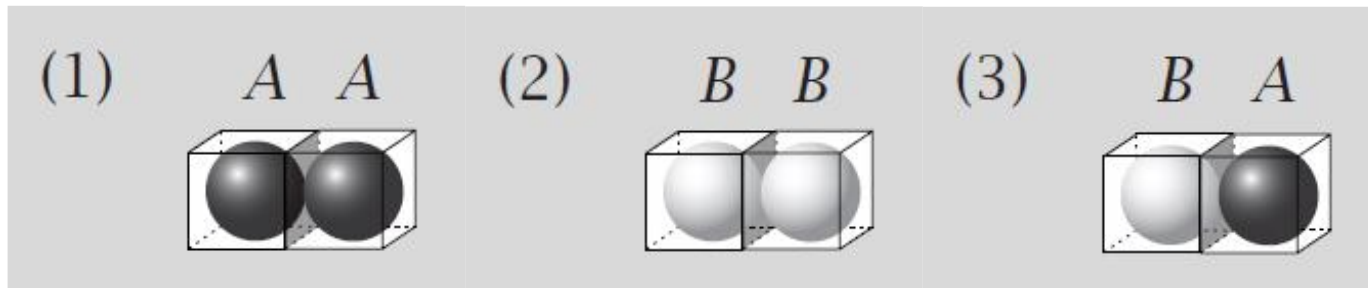
$$\Delta S_{\text{mix}} = M_A \Delta S_A + M_B \Delta S_B = -k_B (M_A \ln \phi_A + M_B \ln \phi_B)$$

$$\Delta S_{\text{mix}} = -M k_B \left(\frac{\phi_A}{N_A} \ln \phi_A + \frac{\phi_B}{N_B} \ln \phi_B \right)$$

Thus, per lattice site,

$$\Delta \bar{S}_{\text{mix}} = -k_B \left(\frac{\phi_A}{N_A} \ln \phi_A + \frac{\phi_B}{N_B} \ln \phi_B \right)$$

Energy of Mixture



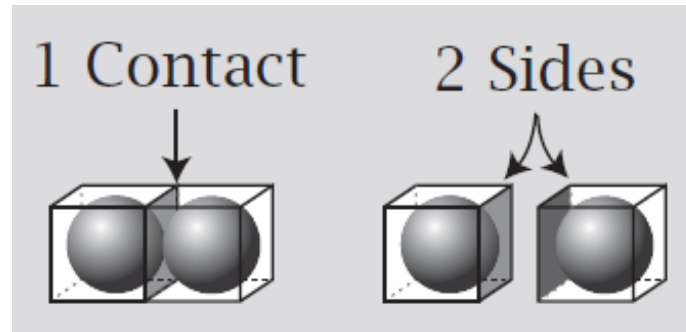
Considering three types of contacts,

$$U = m_{AA}w_{AA} + m_{BB}w_{BB} + m_{AB}w_{AB}$$

where m_{ij} is the number of i - j contacts, and

w_{ij} is the energy of each i - j contact

Contacts and Sides



- Each contact contributes 2 sides
- Each monomer contributes z sides
- Total number of “A” sides = $zM_A N_A = 2m_{AA} + m_{AB}$
- Total number of “B” sides = $zM_B N_B = 2m_{BB} + m_{AB}$

Energy of Mixture

$$zM_A N_A = 2m_{AA} + m_{AB}, \quad zM_B N_B = 2m_{BB} + m_{AB}$$

Rearrangement leads to

$$m_{AA} = (zM_A N_A - m_{AB})/2$$

$$m_{BB} = (zM_B N_B - m_{AB})/2$$

which gives

$$U = \frac{zM_A N_A - m_{AB}}{2} w_{AA} + \frac{zM_B N_B - m_{AB}}{2} w_{BB} + m_{AB} w_{AB}$$

$$U = \frac{zw_{AA}}{2} M_A N_A + \frac{zw_{BB}}{2} M_B N_B + \left(w_{AB} - \frac{w_{AA} + w_{BB}}{2} \right) m_{AB}$$

What is m_{AB} ?

The probability to have B monomer in any site:

$$p_B = M_B N_B / M$$

The average number of AB contacts that each A monomer makes:

$$z p_B = z M_B N_B / M$$

The total number of AB contacts:

$$m_{AB} \approx M_A N_A \times (z M_B N_B / M) = z M_A M_B N_A N_B / M$$

Energy of Mixture

$$U = \frac{z w_{AA}}{2} M_A N_A + \frac{z w_{BB}}{2} M_B N_B + \left(w_{AB} - \frac{w_{AA} + w_{BB}}{2} \right) m_{AB}$$

$$m_{AB} = z M_A M_B N_A N_B / M$$

$$\bar{U} = \frac{z w_{AA}}{2} \phi_A + \frac{z w_{BB}}{2} \phi_B + \chi k_B T \phi_A \phi_B$$

where the **exchange parameter** χ is

$$\chi = \frac{z}{k_B T} \left(w_{AB} - \frac{w_{AA} + w_{BB}}{2} \right)$$

Energy of Mixing

$$\bar{U} = \frac{ZW_{AA}}{2} \phi_A + \frac{ZW_{BB}}{2} \phi_B + \chi k_B T \phi_A \phi_B$$

- Pure A: $\phi_B = 0$

$$\bar{U}_A = \frac{ZW_{AA}}{2} \phi_A$$

- Pure B: $\phi_A = 0$

$$\bar{U}_B = \frac{ZW_{BB}}{2} \phi_B$$

$$\Delta \bar{U}_{\text{mix}} = \bar{U}_{A+B} - \bar{U}_A - \bar{U}_B = \chi k_B T \phi_A \phi_B$$

Flory-Huggins Model

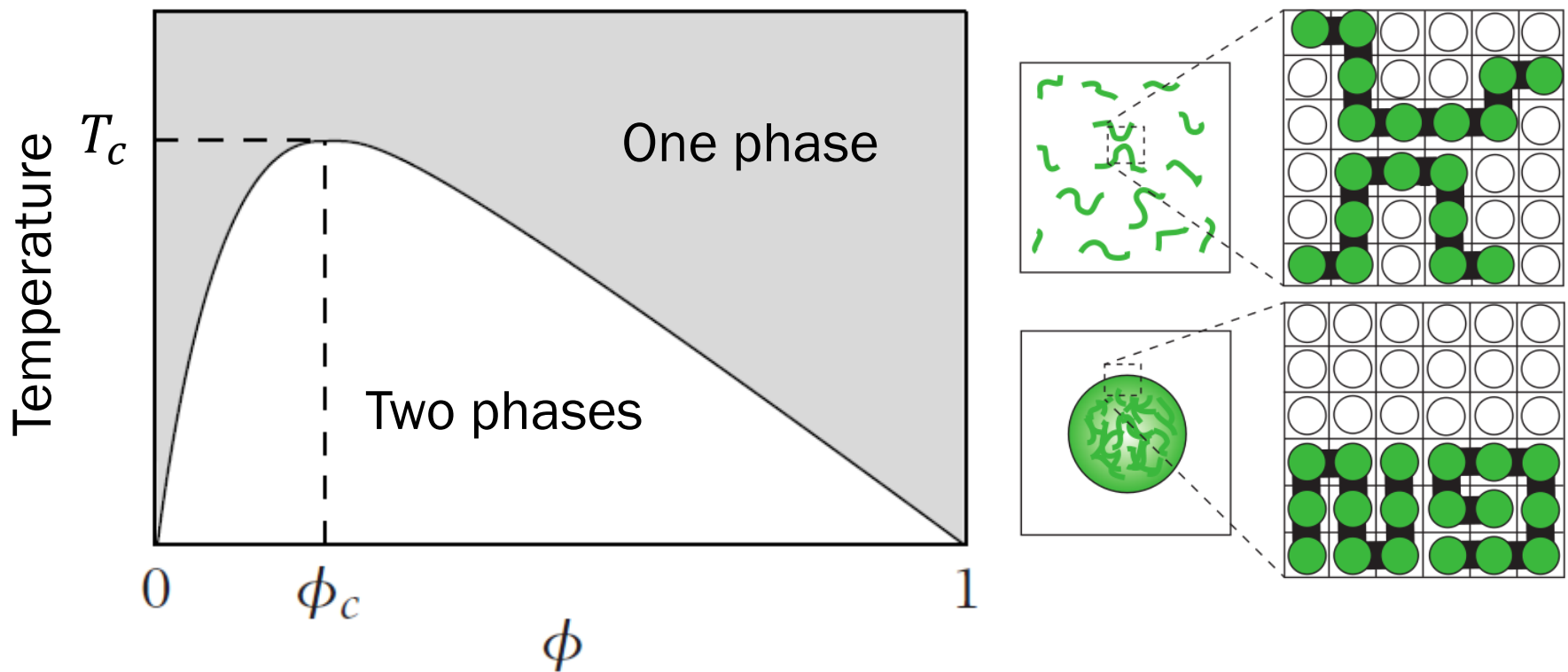
$$\Delta \bar{S}_{\text{mix}} = -k_B \left(\frac{\phi_A}{N_A} \ln \phi_A + \frac{\phi_B}{N_B} \ln \phi_B \right)$$

$$\Delta \bar{U}_{\text{mix}} = \chi k_B T \phi_A \phi_B$$

$$\Delta \bar{A}_{\text{mix}} = \Delta \bar{U}_{\text{mix}} - T \Delta \bar{S}_{\text{mix}}$$

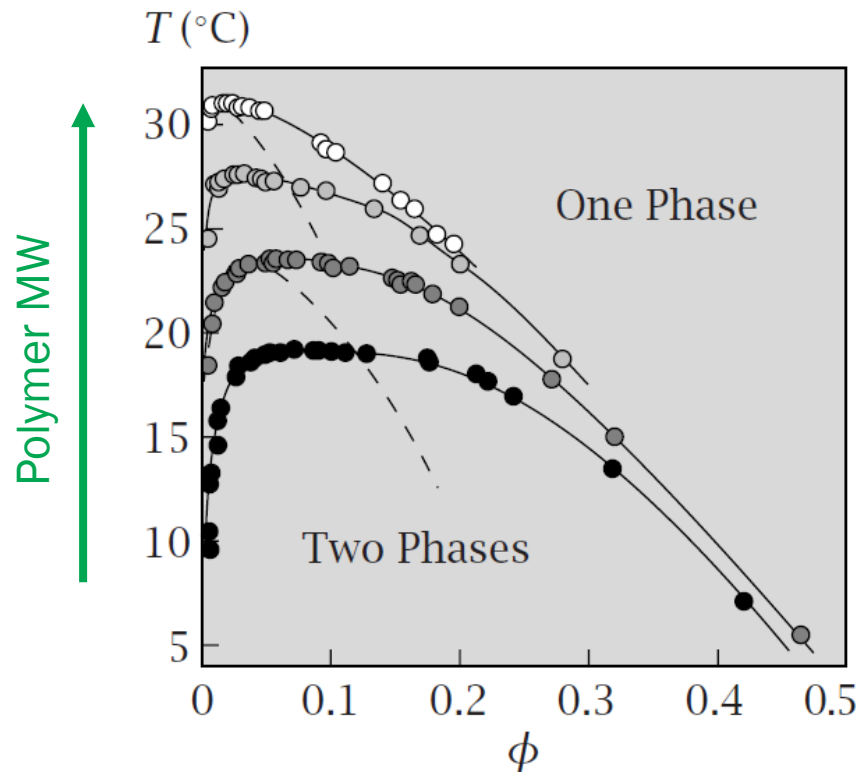
$$\Delta \bar{A}_{\text{mix}} = k_B T \left(\frac{\phi_A}{N_A} \ln \phi_A + \frac{\phi_B}{N_B} \ln \phi_B + \chi \phi_A \phi_B \right)$$

Phase Diagram



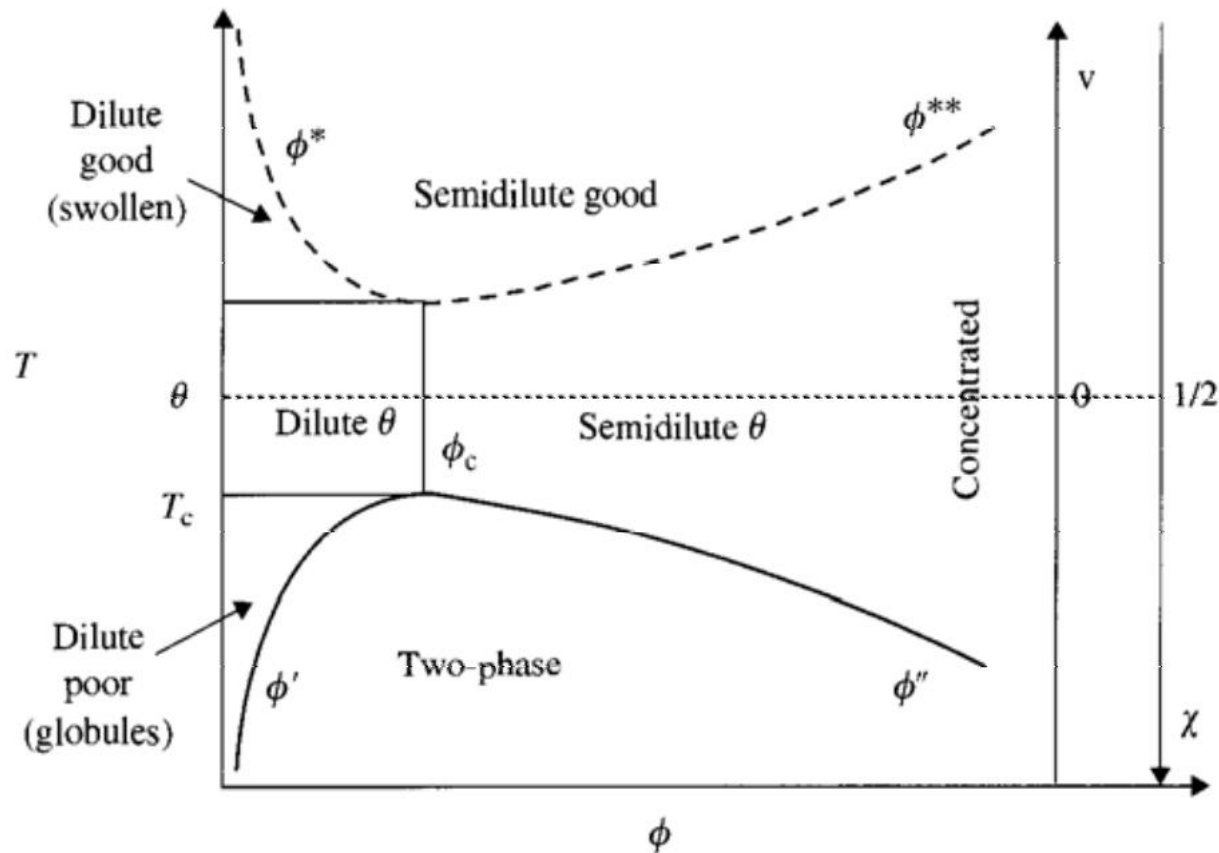
(Images: Dill and Bromberg, *Molecular Driving Forces*, 2nd Ed., Figure 32.7;
Brangwynne et al., *Nature Physics* (2015))

Polystyrene in Cyclohexane



(Image: Dill and Bromberg, *Molecular Driving Forces*, 2nd Ed., Figure 32.8)

Summary



(Image: Rubinstein and Colby, *Polymer Physics* (2003), Fig. 5.1)